

There are two things that hold true when it comes to artificial intelligence. One. It's everywhere.

My toothbrush just got an AI update this week.

And two. The field is changing rapidly, making it hard to keep up even for those of us who work in tech.

So, I've put together my top seven AI terms

that I think are important to be familiar with as AI continues to progress. How many do you already know?

Well, let's find out.

And I'm gonna to start at number one with something that I'm quite sure that you have heard of.

And that's agentic AI.

Everybody and their grandmother seems to be building the next generation of AI agents.

But what exactly are they?

Well, AI agents, they can reason and act autonomously to achieve goals. Something like a chatbot that only responds one prompt at a time,

AI agents, they run autonomously.

They go through a number of stages.

So, first of all, they perceive their environment.

Once they've done that, they move on to a reasoning stage, and that's where they look to see what the next best steps forwards are.

Then they move on to act on the plan that it's built through the reasoning, and then observes the results of that action.

And around and around we go.

Now agents can work as well in all sorts of roles.

They could be your travel agent to book a trip.

They could be your data

analyst to spot trends in quarterly reports.

Or they could perform the role of a DevOps engineer, detecting anomalies in logs and spinning up containers

to test fixes and rolling back faulty deployments.

And AI agents are typically built

using a particular form of large language models,

and that is known as number

two, large reasoning models.

Now these are specialized LLMs that have undergone reasoning-focused fine tuning. So unlike regular

LLMs that generate responses immediately,

reasoning models, they're trained to work

through problems step by step,

which is exactly what agents need

when planning complex, multistep tasks.

Now, the reasoning model is trained on problems with verifiably correct answers.

So math problems or code that can be tested by compilers and through reinforcement learning, the model learns to generate reasoning sequences that lead to correct final answers. So, every time you see a chatbot pause before it responds back to you by saying, thinking. Well, that's the reasoning model at work, generating an internal chain of thought to break down a problem step by step before generating a response. Now let's get a bit lower level and talk about number three, which is vector database. So, in a vector database, we don't store raw data like text files and like images just as blobs of data. We actually use something called an embedding model. And that embedding model is used to convert that data from these images here into actually a vector. Now, what is a vector? Well, a vector is essentially just kind of a long list of numbers. And that long list of numbers captures the semantic meaning of the context. Now, what's the benefit of doing that? Well, in a vector database, we can perform searches as mathematical operations, looking for vector embeddings that are close to each other. And that translates to finding semantically similar content. So, we might start, with let's say a picture of a mountain vista. Something like this. And then that picture is broken down by the embedding model into vectors, a multidimensional numeric vector. And we can perform a similarity search to find items that are similar to that mountain picture by finding the closest vectors in the embedding space. Or it could be similar text articles, or it could be similar music files. Whatever you want. Now vector databases, they play a big role in implementing number four. And that is RAG or retrieval augmented generation. Now, RAG makes use of these vector databases. And it uses it to enrich prompts to an LLM. So, we start here with a RAG retriever component.

Now that might take in an input prompt from a user.

And it's going to turn it into a vector using an embedding model.

That's the thing that ties it into that series of numbers.

And then, once we've done that, we can perform a similarity search in the vector database.

Now that vector database will return something, and we'll return that all the way back to the large language model prompt that we started with.

And we'll embed into that prompt now the stuff that came out of that vector database.

So, I can I can ask a question about let's say company policy.

And then this RAG system is going to pull the relevant section from the employee handbook to include in the prompt.

Now, number five,

that Model Context Protocol or MCP.

This is a really exciting one because for large language models to be truly useful, they need to interact with external data sources and services and tools.

And MCP standardizes how applications provide context to LLMs.

So, if you want your large language model here to be able to connect to stuff.

Perhaps we want to connect to an external database, or maybe we want to go to some kind of code repository, or maybe even to an email server, or really any kind of external system.

Well, MCP makes that connection standardized. So, instead of developers having to build one off connections for each new tool,

MCP provides a standardized way for AI to access your systems.

So basically we have here an MCP server.

And that is how the AI knows exactly what to do to get through to any one of these tools.

It connects through that MCP server connection. Okay.

Now, for number six.

That's a mixture of experts or MOE.

And we've had the idea of MOE for a good while, actually, since the paper was published in a scientific journal in 1991.

But, it's never been more relevant than it is today.

You see, MoE divides a large language model into a series of experts.

I'm just gonna to draw three, but there could be 100 plus of these.

These are specialized neural subnetworks.

And then it uses a routing mechanism to activate only the agents it needs for a particular task.

or only the experts in this case that it needs for a task.
And then, well, then it's going to perform a merge process.
So, because we activated these two experts,
we'll merge these two.
And this performs mathematical operations
to combine the output
from these different experts into a single representation
that continues through the rest of the model.
And it's a really efficient way to scale up model size
without proportional increases in compute costs.
So, for example, MoE models, like IBM Granite's 4.0 series,
that can have dozens of different experts here. But for any given token,
it will only activate these specific experts it needs.
And that means, though, while the whole model
might have billions of total parameters,
it only uses a fraction of those active parameters at inference time.
And look, for number seven,
I'm gonna throw in a big one, ASI,
artificial superintelligence.
It's the goal of all the frontier AI labs.
And at this point, it is purely theoretical.
It doesn't actually exist and we don't know if it ever will.
Now, today's best models,
they're slowly approaching a different standard, which is AGI.
That's artificial general intelligence.
Now that's also theoretical. But,
if realized, AGI will be able to complete
all cognitive tasks as well as any human expert.
ASI is one step beyond that.
So, ASI systems would have an intellectual scope
beyond human level intelligence,
potentially capable of recursive self-improvement.
So, basically an ASI system could redesign and upgrade itself,
becoming ever smarter in an endless cycle.
It's the kind of development that would either solve humanity's biggest problems
or create entirely new ones that we can't even imagine yet.
And if that happens, well,
I think it's probably a pretty good idea
that we keep the term ASI on our radar.
So, that's my seven.
But I'm curious, what's the AI term
you think that should have made it onto this list?
Let me know in the comments.
Hey, Martin. Hey Graeme.
Hey, this is really cool stuff.
This AI and these terms. Fascinating.

Yeah, and I came up with seven, but I could have come up with 70.
There's so much going on in this space. I I I bet you could.
And you know what? There is so much going on.
We are actually going to be talking about AI a lot at the IBM
TechXchange conference in Orlando this October.
And you know what? I'm gonna be there as well.
I know it's gonna be so exciting!
There's going to be so much going on.
We are going to have. Let's see.
We're gonna have boot camps.
We're going to have workshops. There's going to be sessions.
There's going to be live demos, certifications, all kinds of things going on.
So much more when it comes to AI.
But but Martin, what are you going to be doing there?
Well, I'm going to be bringing my light board pens
and this light board to the sessions as well.
Oh my God! It's so exciting! I'm so excited have you there!
Yes. So we're actually going to have a light board studio set up.
And ah we're going to be performing light boards live.
So if you always wondered how do I write backwards,
you're going to find out in person at the event.
And da also we'll be kind of teaching
how to perform a light board
video yourself. The sort of things that you need to know for that.
Wow. So you get to meet a celebrity and maybe become one yourself.
That sounds really exciting.
I can't wait to welcome you down to Orlando.
It's just going to be a blast. So looking forward to it. Can't wait.
All right. Hope we see you there too.
So, go to ibm.com/techXchange and we'll see you down there.